

Aspect shapes the meaning of attitude reports

(De-)composition & modification of attitude predicates workshop

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Introduction

Attitude reports also describe eventualities

- | | | |
|-----|--|----------------|
| (1) | Nadine believes that I'm in Konstanz. | state |
| (2) | a. Travis is imagining that it's his birthday. | activity |
| | b. Prerna found out that I'm in Bochum. | achievement |
| | c. Julian learned that math was hard. | accomplishment |

Eventuality-related (=aspectual) properties of attitude predicates:

- ✧ are not well catalogued, stative privilege?
- ✧ interact with their modal and combinatorial properties.

Not well catalogued

[S]entential complements will figure prominently in any full description of the properties of [verbs relating to communication and the transfer of ideas], but a systematic treatment of such complements is outside the scope of this book.

Levin (1993: p. 202) *English Verb Classes and Alternations*

Interact with combinatorial and modal properties

Aspect and interrogative embedding

Big question:

What clause types do different attitude predicates combine with?

Partial answer:

to be refined

Dynamic ‘think’ combines with interrogatives.

Stative ‘think’ resists them.

- (3) a. #Alice thinks whether she should invite Tom.
- b. Alice was thinking whether she should invite Tom.

See also Grohne (2016), White & Rawlins (2018), Özyıldız (2021), White (2021).

Interact with combinatorial and modal properties

Aspect and neg-raising

Big question:

What inferences do attitude predicates license qua modals?

Partial answer:

to be refined

Stative ‘think’ licenses the neg-raising inference.

- (4) Alice doesn’t think that she should invite Tom.
 $\rightsquigarrow$ Alice thinks that she shouldn’t invite Tom.

Dynamic ‘think’ doesn’t.

- (5) Alice isn’t thinking that she should invite Tom.
 $\nrightarrow$ Alice is thinking that she shouldn’t invite Tom.

For neg-raising, see Bervoets (2014, 2020), Xiang (2020), Jeretič & Özyıldız (2022).
For telicity $\sim$ exhaustivity, Grohne (2017); For perfectivity $\sim$ factivity, Zuchewicz (2020); a.o.

So what?

A partial answer

A long standing program

The semantic properties of attitude predicates explain their combinatorial properties.

“Think” doesn’t combine with interrogatives because it is neg-raising.

Challenges

“Think” (e.g.,) is attested with interrogatives, so our available...

- ✧ *generalizations are false.*
- ✧ *theories are wrong.*

So what?

A partial answer

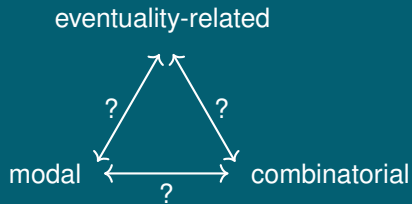
Authors have mostly relied on modal properties of attitude predicates as explanatory factors.
(Neg-raising, but also veridicality, preferentiality, etc.)

Incorporating aspect into the mix...

- ✧ Provides new answers to old questions.
- ✧ Raises new questions, for which we have old answers.

More concretely, let's:

- | | |
|--|---------------|
| ✧ better characterize the behavior of <i>think</i> , | new answers |
| ✧ appeal to familiar tools to model it, | old tools |
| ✧ and speculate about other predicates. | new questions |



A long standing program + challenges

A familiar paradigm

Predicates vary in whether they may combine with declaratives & interrogatives.

- | | | | |
|-----|-----------------|---------------------------|-----------------|
| (6) | Nadine believes | that I'm in Konstanz. | |
| | | *whether I'm in Konstanz. | "anti-rogative" |
| (7) | Nadine knows | that I'm in Konstanz. | |
| | | whether I'm in Konstanz. | "responsive" |
| (8) | Nadine wonders | *that I'm in Konstanz. | |
| | | whether I'm in Konstanz. | "rogative" |

There *is* something to explain

*What bothers us is WHY, e.g., we have ‘believe that’ but not ‘*believe who’, even though the latter looks as if it ought to make sense (cf. ‘have an opinion as to who’).*

Boër (1978: p. 312)

A common program

Grimshaw 1979, Zuber 1982, ...

The semantic properties of attitude predicates' explain their combinatorial properties.

Semantic Restrictions on Certain Complementizers

R. Zuber

CNRS, Paris

Any researcher interested in the relation between linguistic form and its semantic content will find quite natural the question of whether or not the absence of some particular linguistic form can be explained on semantic grounds. In this note I will show that the impossibility of a *wh*-complementizer after *neg*-transportation verbs has semantic justification.

Focus on the neg-raising generalization

Zuber 1982, Egré 2008, Mayr 2019, Theiler et al. 2019

- (9) If a predicate is neg-raising, it doesn't combine with interrogatives.

- (10) **Neg-raising**
Nadine doesn't think that I'm in Konstanz.
↪ Nadine thinks that I'm not in Konstanz.

- (11) **No interrogatives (?)**
*Nadine doesn't think whether I'm in Konstanz.

‘Think’ and question embedding

Unacceptable sentences support the neg-raising generalization

- (12)
- a. *John thought who Bill saw. (Grimshaw 1979)
 - b. *John thought why Mary left. (Rawlins 2013)
 - c. *I thought who will be invited to the party. (Dayal 2016)
 - d. *John thinks whether Mary drinks. (Mayr 2019)

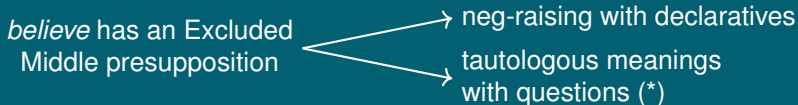
An explanation of the neg-raising generalization

Meaning driven unacceptability

Mayr 2019, Theiler et al. 2019

- (9) If a predicate is neg-raising, it doesn't combine with interrogatives.

because



An explanation of the neg-raising generalization

Meaning driven unacceptability

Mayr 2019, Theiler et al. 2019

Assumptions about truth conditions

(13)	$\llbracket x \text{ believe whether } p \rrbracket$	
	defined only if	$B_x p \vee B_x \neg p$ EM + Q semantics
	true iff	$B_x p \vee B_x \neg p$ Q semantics

presupposition entails assertion $\Rightarrow$ true whenever defined

Architectural assumptions

Gajewski 2007, a.o.

Sentences with tautologous or contradictory meanings are ungrammatical

Hence

Neg-raising predicates are ungrammatical with embedded questions.

Other generalizations

not necessarily all true (together)

... of **damnation**

- (9) If a predicate is neg-raising, it doesn't combine with interrogatives.
- (14) If a predicate is non-veridical and preferential, it doesn't combine with interrogatives. Uegaki & Sudo (2019)
⇒ *hope wh-

... of **salvation**

- (15) If (and only if) a predicate is veridical with declaratives, it also combines with interrogatives. Égré (2008)
⇒ know that/wh-

‘Think’ and question embedding

Acceptable sentences challenge the neg-raising generalization

- (16) People **will have to think whether** they want four more years of that.
- (17) I’m **trying to think whether** I’d have been a star today or not.
- (18) Often, when listening to other players, I **start to think whether** there’s an unwritten law for guitarists to never play an interval bigger than the major third.
- (19) I **was thinking whether** there was a way to help more than one person.
- (20) Jo sat down and **thought whether** she should invite Bo.

Exx. (16)–(19) from White (2021).

Spot check

Assume a version of “think” devoid of an EM presupposition:

$$(21) \quad \begin{aligned} \llbracket \text{think}_{-EM} \rrbracket &= \lambda Q_{\langle \langle s,t \rangle, t \rangle} \lambda x_e. \exists p \in Q : B_x p \\ \llbracket \text{whether Jo invites Bo} \rrbracket &= \{ \text{invite}(j, b), \neg \text{invite}(j, b) \} \end{aligned}$$

This predicts:

$$(22) \quad \begin{aligned} &\text{Jo thought whether she should invite Bo.} \\ &\exists p \in \{ \text{invite}(j, b), \neg \text{invite}(j, b) \} : B_x p \end{aligned} \quad \text{non !}$$

⇒ Not triggering the EM presupposition. . .

- ✧ might help with selection, but
- ✧ fails to assign (22) a deliberative meaning.

assume no ambiguity

What aspect reveals about question embedding and neg-raising

Core generalization about question embedding

The kind of eventuality description introduced by 'think' depends, in part, on the type of its sentential complement:

- ▶ 'Think that' may describe a state or an event.
- ▶ 'Think wh-' must describe an event.

	think that	think wh-
stative	✓	—
dynamic	✓	✓

Diagnostics I: Ongoing interpretations

- ▶ Stative predicates may describe an ongoing state in the present simple, not in the present progressive.

(23) Dani likes/#is liking snails

- ▶ Dynamic predicates may describe an ongoing event in the present progressive, not in the present simple.

(24) Kim #examines/is examining a patient.

- ▶ Dynamic predicates in the present simple receive 'special interpretations.'

(25) a. Kim examines a patient (in the mornings).
b. Kim walks in. She examines a patient. She walks out.

Diagnostics I: 'Think that' can be stative or dynamic

'Think that p' can describe. . .

- ▶ an ongoing state in the present simple

(26) Alice thinks that she should invite Tom.

- ▶ an ongoing event in the present progressive

(27) Alice is thinking that she should invite Tom.

✓ actively, worriedly, etc.

'Think that p' has both interpretations, it passes both tests.

Diagnostics I: 'Think wh-' must be dynamic

'Think wh-'

- ▶ can describe an ongoing event in the present progressive

(28) Alice is thinking whether she should invite Tom.

- ▶ can't describe an ongoing state in the present simple

(29) #Alice thinks whether she should invite Tom.

Diagnostics I: 'Think wh-' must be dynamic

(29) #Alice thinks whether she should invite Tom.

- ▶ This judgment is a # and not a *
- ▶ The sentence does have
 - ▶ a habitual reading:
Alice usually thinks whether she should invite Tom.
 - ▶ a play by play reading:
Alice pops in, thinks whether she should invite Tom,
and pops out.

This is further evidence for the eventivity of 'think wh-'

- ▶ The sentence can't mean *know* or *have a thought about whether p*

Back to acceptable/unacceptable frames

Like other eventive predicates *think wh-* is odd in the present simple:

(30) *John thinks whether Mary drinks.

It is acceptable in frames that tolerate eventives:

(31) I'm trying to think whether I'd have been a star today or not.

Our core generalization covers (most of) these cases.
There are wrinkles, in the appendix.

Core generalization about *think* and neg-raising

Does the fact that *think* embeds questions falsify (32)?

(32) If a predicate is neg-raising, it doesn't combine with interrogatives.

The answer is a nuanced “no”

- ▶ Neg-raising & question embedding do exclude each other.
- ▶ But this is because of a competing requirement:

Question embedding requires eventivity.

Neg-raising requires stativity. ← to be argued for now

- ▶ Not because neg-raising causes anti-rogativity.

Empirical arguments in favor of “Neg-raising $\Rightarrow$ stative”

✧ B and B' are not synonymous

(33) A: What's going on over there?

B #Alice isn't thinking that she should invite Tom.

B' Alice is thinking that she shouldn't invite Tom.

✧ Strict NPIs are not licensed by matrix negation

(34) *When the barking dog startled him, the farmer wasn't thinking that the tree had fallen until late last night.

Can't mean:

... the farmer was thinking that the tree hadn't fallen until late last night.

Bervoets (2014)

Disclaimer: The data here is also tricky.

Disclaimer: Showing this is tricky

Being in the progressive doesn't force a predicate to be dynamic.

✧ White does the same exercise as us and reaches a different conclusion:

(35) I wasn't thinking there was a way to help more than one person (at a time) until Jo got back from lunch.

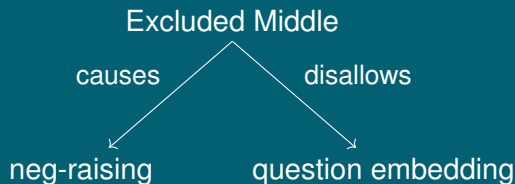
✧ Similar examples are attested elsewhere

(36) Coach Pete DeBoer says he's not thinking that the playoff dreams are over.
"I don't feel that way [. . .]"

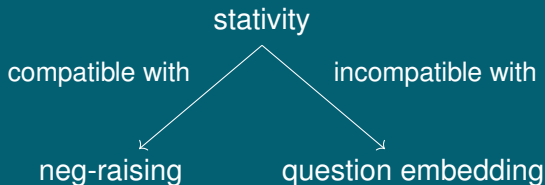
Suggestion: These are instances of stative progressives.

The resulting picture (programmatic)

Out:



In:



Relating the stative and dynamic guises of alternating predicates

“Think” alternates between a stative and a dynamic guise.

How to relate them in such a way that we. . .

- ✧ capture their truth conditions,
- ✧ only derive neg-raising for stative *think that*.
- ✧ avoid assigning *think wh*- ongoing state readings,

A logical space

Assume that one of stative or dynamic is derived from the other.

Two hypotheses appealing to familiar devices:

H1 Stative derived from dynamic

- ✧ *Think* describes events.
- ✧ Stative thought reports are generics of eventive ones.

Krifka et al. (1995) , a.o.

H2 Dynamic derived from stative

- ✧ *Think* describes states.
- ✧ Eventive thought reports derived via aspectual coercion.

Moens & Steedman (1988), a.o.

Let's see what it takes to implement H1.

Stative Thoughts As Generics (STAG)

Lexical assumptions

Think has the basic denotation of an *eventive* predicate:

$$(37) \quad \llbracket \text{think} \rrbracket = \lambda Q_{\langle \langle s, t \rangle, t \rangle} \lambda x_e \lambda e_v. \text{think}'(e, x, Q)$$

Immediate consequence:

Dynamic *think that* and *think wh-* are derived.

To do:

- ✧ Derive stative *think that*, with the right truth conditions and neg-raising,
- ✧ without deriving stative *think wh-*.

STAG

Stative *think*: Target truth conditions

(38) Alice thinks that it's raining.

⇔ If you were to ask Alice whether it's raining, she would
eventively think that it is. *in ordinary circumstances*

Left to the audience to find: Two ways of breaking the equivalence.



STAG

Eventive $\sim$ stative doublets similar to *think that*

- (39) a. Alice is speaking French.
 b. Alice speaks French.
 $\approx$ Alice is able to (eventively) speak French.
- (40) This machine crushes oranges.
 $\approx$ This machine is designed to (eventively) crush oranges.

Putting the AG in STAG

Stative member of pair derived via the application of GEN.

- (41) a. [GEN [this machine crush oranges]]
b. $\forall w$: this machine performs the action for which it was designed at $w \rightarrow$ it crushes oranges at w .

The **restrictor** of GEN is determined contextually.

Assume the same for stative *think that*:

- (42) Alice thinks that it's raining.
a. [GEN [Alice think that it's raining]]
b. $\forall w$: Alice is asked if it's raining at $w \rightarrow$ she eventually thinks that it's raining at w .

A shorthand

$\forall w$: Alice is asked if it's raining at $w \rightarrow$ she eventually thinks that it's raining at w .

restrictor worlds:

- *asked*($a, ?r$)

are a subset of scope worlds:

- *think*(a, r)

(How) can STAG help with neg-raising?

Ingredient 1: The scope of negation

With some generics, negation is interpreted in GEN's scope.

- (43) Alice doesn't smoke.
- ✓ Usually, it not the case that Alice smokes.
 - ✗ It's not the case that Alice usually smokes.

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In (44), let's also interpret negation GEN's **scope**.

- (44) Alice doesn't think that it's raining.
 $\forall w$: Alice is asked if it's raining at $w \rightarrow$ **she doesn't
eventively think that it's raining at w .**

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 $\forall w$: Alice is asked if it's raining at $w \rightarrow$ **she doesn't
eventively think that it's raining at w .**

This is not enough to derive neg-raising.

(How) can STAG help with neg-raising?

Ingredient 2: Asking questions

Pragmatic principle:

If S is asked a question, it is also assumed that she believes an answer and that she will think ($\approx$ say) the answer she believes.

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Ingredient 2: Asking questions

Pragmatic principle:

If S is asked a question, it is also assumed that she believes an answer and that she will think ($\approx$ say) the answer she believes.

This enriches GEN's restrictor into:

restrictor worlds:

- $asked(a, ?r)$
- $B_a(r) \vee B_a(\neg r)$
- $B_a(r) \rightarrow think(a, r) \wedge B_a(\neg r) \rightarrow think(a, \neg r)$

are a subset of scope worlds:

- $\neg think(a, r)$

(How) can STAG help with neg-raising?

Ingredient 2: Deriving the inference

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conclusion:

- $\neg B_a(r)$

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conclusion:

- $\neg B_a(r)$
- $B_a(\neg r)$

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are a subset of scope worlds:

- $\neg think(a, r)$

conclusion:

- $\neg B_a(r)$
- $B_a(\neg r)$
- $think(a, \neg r)$

We're allowed to do this because all restrictor worlds are $\neg think(a, r)$ worlds.

How does STAG help with neg-raising?

We've successfully strengthened the consequent of (45):

(44) Alice doesn't think that it's raining.

$\forall w$: Alice is asked if it's raining at $w \rightarrow$ she doesn't
eventively think that it's raining at w .

Into:

(45) $\forall w$: Alice is asked if it's raining at $w \rightarrow$ she eventively
thinks that it's **not** raining at w .

The missing ongoing state reading for *think wh-*

Think wh- has an ignorance entailment: when atelic

- (46) Alice is thinking whether it's raining.
⇒ Alice neither believes that it's raining nor that it's not.

Assume that when we try to say:

- (47) #Alice thinks whether it's raining.

We keep GEN's restrictor from before and add the ignorance entailment to its scope.

The missing ongoing state reading for *think wh-*

restrictor worlds:

- $asked(a, ?r)$
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are a subset of scope worlds:

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contradiction

The missing ongoing state reading for *think wh-*

This is a contradiction:

(47) #Alice thinks whether it's raining.

Further work: What about “Alice doesn't think whether it's raining”? (Currently not obviously a contradiction unless ignorance inference may sometimes project.

Spot check

Present simple *think wh-* isn't ungrammatical!

- ✧ It's # under a “Alice is curious whether” reading.
- ✧ There is no “Alice has a belief about whether” reading.

Habitual examples like (46) are possible, but we arguably have **different restrictors** there (or HAB instead of GEN).

- (46) Alice thinks whether she should invite Tom.
Every **time she throws a party**, Alice (eventively) thinks whether she should invite Tom.

What have we achieved, and how

With an eventive, non-picky lexical entry for *think* PLUS

- ✧ GEN

- ⇒ Intuitive truth conditions for *x thinks that*

- ✧ scope of negation & pragmatics of asking Qs

- ⇒ The neg-raising inference for *x doesn't think that*

- ✧ *think wh*-s ignorance inference

- ⇒ No spurious ongoing state readings

Popping back out

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But *which* semantic properties are explanatory?

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By placing *think* under the microscope, we have seen that its aspectual properties were crucial in describing and explaining its behavior.

We have also seen that it is feasible to model known as well as yet-to-be-understood aspects of their behavior in a way that is eventuality-aware.

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What about other classes of predicates? Other languages?
Other aspect-based empirical generalizations?

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Semantic properties explain combinatorial properties.

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What about other classes of predicates? Other languages?
Other aspect-based empirical generalizations?

What about H2, deriving eventive reports from a stative core via aspectual coercion?

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Most immediately:

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Appendix

Wrinkle 2 with the data

Uses of *think* that are close to *stative* and *question embedding*.

- (49) When asked whether she thought whether she may have come across as slightly obsessive, Becky replied, 'No.'

Situation described by (49):

- (50) A asks B: "Do you think that you may have come across as obsessive?"
- a. ✓ What is your opinion?
 - b. × Do you engage in thinking that?

Similarly:

- (51) I was thinking whether we should leave.
≈ I think that we should leave.

Wrinkle 2 with the data

My reactions:

- ▶ Do these examples replicate with kinds of embedded questions other than with *whether*?
- ▶ Present simple “think whether” is still odd in these frames.

- (52)
- a. #I think whether I may have come across as obsessive.
 - b. #I think whether we should leave.

- ▶ Highlighting?
Euphemism?
Complementizer agreement??

More on H2

In relating stative and dynamic *think that*, we have two options:

- ▶ Derive dynamic *think* from stative *think*.
- ▶ Derive stative *think* from dynamic *think*.

The stative to dynamic hypothesis is independently motivated for certain attitude predicates:

- (53) Aspectual coercion
?She was suddenly knowing all of the answers.

Given that there is stative *think*, coercion might even apply to it.
But is one argument in favor of dynamic to stative.

Meaning neutralization vs. generation

Many eventive verbs have stative counterparts whose at issue component is just “believe.”

(54) Alice {imagines, supposes, agrees, remembers, ...}
that it's raining.

- ▶ It is easy to efface layers of meaning in a unified way.
GEN push these layers to non-actual worlds.
- ▶ It is harder to generate layers of meaning in a unified way.

Meaning neutralization vs. generation

It is unclear what stative ‘imagine’ would mean: (55)?

$$(55) \quad \llbracket \text{imagine} \rrbracket = \lambda Q. \lambda x. \lambda s. \text{dox}(s, x) \in Q$$

Operators OCC that would derive occurrent attitudes from, e.g., (55), would be defined on a case by case basis.

$$(56) \quad \llbracket \text{OCC}_{\text{imagine}} \rrbracket = \\ \lambda V. \lambda Q. \lambda x. \lambda e. \exists s : V(Q)(x)(s) \wedge \text{imagine}'(e, x, Q)$$

More fatal:

If dynamic “imagine” derived from stative/belief-imagine via OCC, we would expect it to imply belief.

$$(57) \quad \text{Travis is imagining that dragons are invading Michigan.} \\ \nrightarrow \text{Travis believes that dragons are invading Michigan.}$$

Diagnostics II: Narrative time advancement

In a narrative sequence in the simple past,

- ▶ stative predicates provide background information.

(58) Dani arrived at the restaurant.

She **knew the menu.**

She ordered the snails. [2 things happened]

- ▶ dynamic predicates move narrative time forward.

(59) Dani arrived at the restaurant.

She **read the menu.**

She ordered the snails. [3 things happened]

English statives in the simple past are imperfective, but dynamic predicates, perfective.

Diagnostics II: 'Think that' can describe a state or an event

In the simple past, 'think that p' may provide background information or move narrative time forward.

- (60) Dani arrived at the restaurant.
She thought that she should have something fun.
She ordered the snails.
[ambiguous: 2 things happened or 3]



Diagnostics II: 'Think wh-' must describe an event

In the simple past, 'think wh-' cannot provide background information and must move narrative time forward.

- (61) Dani arrived at the restaurant.
She thought whether she should have something fun.
She ordered the snails. [3 things happened]

